

Business Case and Business Rules – Menciau's Chopstix Order Management System

Oct 2, 2026 · @QB · Team 9, Fusion Solutions Experience (XP) · CIS 4375

Purpose

This document gives Noor what she needs to draw the scope diagram (task T03, due 5 Oct): the business case, the system boundary, the external entities and data flows, and the business rules.

Everything here comes from the Project Charter, the Requirements Traceability Matrix (RTM, 2 Oct draft), and the team's as-is and to-be swim lanes and SIPOC diagrams. Business rules are numbered BR-01 onward and cite the RTM IDs they come from. The open questions at the end need answers before scope freezes.

Business case

The recommended option is a custom online pickup ordering and kitchen ticketing system. It removes the counter line and the verbal hand-off to the cook, and gives the owner order history and sales data for the first time, at no build cost to the restaurant.

Current state (as-is)

Every order is taken at the counter today. The customer waits in line and tells the cashier the order, the cashier enters it into the POS and takes cash or card, then relays the order to the cook by voice. The cook prepares the food, which is plated for dine-in or bagged for takeout.

Pain points:

- Customers wait in line at the counter to order and pay.
- Orders reach the cook by voice, so they can be missed or misheard.
- There is no order history or sales report.
- Prices are tracked by hand.
- Customers cannot order ahead or pay before they arrive.

Proposed state (to-be)

1. The customer opens the ordering website, builds a cart, and checks out with name, email, and pickup time.
2. The system checks business hours and lead time, then prices the order from the database.
3. The customer pays through Stripe by card, Apple Pay, or Google Pay.
4. When Stripe confirms payment, the system marks the order paid and shows a confirmation number. A declined payment returns the customer to checkout.
5. The ticket appears on the kitchen screen within 5 seconds.
6. The cook marks the ticket In Progress, prepares the food, then marks it Ready, which emails the customer that the order is ready.
7. The counter bags the order and checks the order number, and the customer picks it up. A no-show stays in Ready until the owner closes or cancels it.

Walk-in orders keep the current counter process. Kitchen and counter staff share one view and one role. The owner dashboard reads every paid online order for history and reports.

Options considered

Option	What it means	Decision
1. Do nothing	Keep counter-only ordering	Rejected: the line, the verbal relay, and the missing data remain
2. Third-party ordering platform	List the menu on an outside ordering or delivery marketplace	Rejected: per-order commissions, the platform controls menu and customer data, and delivery is out of scope
3. Custom order management system	Build this project, owned by the restaurant	Recommended: ordering, kitchen tickets, and reporting share one database

Benefits

Type	Benefit	How it is measured (RTM)
Revenue	A new channel to order ahead, including catering trays paid online	Sales and order count by date range (F28)
Savings	Less staff time relaying orders, fewer missed or wrong orders	Paid order on the kitchen view within 5 seconds (F15)
Customer experience	No line, a chosen pickup time, card or mobile wallet payment	An order takes no more than five screens (1.02)
Information	Order history and best sellers for the first time	Best sellers by quantity and revenue (F29)
Control	The owner changes prices and availability without a developer	Owner changes a price without written instructions (2.07)

Costs

- **Build:** none to the restaurant; the student team builds it.
- **Hosting:** Supabase and Vercel free tiers during the course (RTM 4.08).
- **Ongoing:** Stripe transaction fees per payment, plus any paid hosting or email service tier the owner later chooses.
- **Owner time:** requirements sessions, supplying menu and price data, reviewing the build, and one training session.

Recommendation

Proceed with option 3, the system the Program Manager approved in the Project Charter.

Scope boundary

The system covers online pickup orders from cart to kitchen ticket, plus the owner's dashboard. Walk-in orders, delivery, and everything the restaurant does before cooking stay outside it.

Item	Where it sits	Note
Customer ordering website: menu, cart, guest checkout, pickup time, catering trays	Inside	Charter 7.1; RTM F1 to F9
Payment request and confirmation handling	Inside	The system sends the amount and receives the result; card data never enters (F12)
Kitchen ticket view and status tracking, shared by kitchen and counter staff	Inside	F15 to F17; one view, one role
Order-ready email request	Inside	Marking a ticket Ready asks the email service to notify the customer
Owner dashboard: history, search, menu, prices, availability, reports	Inside	F21 to F30; reports labeled "Online orders only"
Owner closes or cancels an order (status only)	Inside	Keeps reports accurate; covers no-shows
Staff login and role-based access for owner and kitchen	Inside	F18 to F20
Data stores: menu and prices, orders and order lines, staff accounts, logs	Inside	Hosted on Supabase
Settings: business hours, lead times, tax rate	Inside	Held as data, not code (3.07)
Customer, owner, kitchen staff, Stripe, email service	Outside, connected	The five external entities on the diagram
Refunds	Outside, not connected	The owner issues them in Stripe's own dashboard
Walk-in and dine-in orders on the existing POS and card reader	Outside, not connected	Walk-ins keep the current process

Item	Where it sits	Note
Delivery and third-party delivery platforms	Outside, not connected	Charter out-of-scope
Marketing: social media, flyers, ads, map listings	Outside, not connected	Charter out-of-scope
Inventory, ingredients, and food vendors	Outside, not connected	Charter out-of-scope
POS hardware integration, native mobile apps	Outside, not connected	Charter out-of-scope
Customer accounts, loyalty, coupons	Outside, not connected	Guest checkout only (F8)
Limits on online orders per pickup slot	Outside, not connected	Known limitation; post-project enhancement
Live payment activation, long-term support	Outside, not connected	Post-project handover items

Vercel and Supabase are the platform the system runs on. Draw them inside the boundary or leave them off; they are not external entities.

External entities and data flows

Five external entities connect to the system: the customer, the owner, kitchen staff (counter staff included), Stripe, and the email service. Each row below is one entity; each flow becomes a labeled arrow on the diagram.

Entity	Flows into the system	Flows out of the system
Customer	Cart items, sizes, spice choice, special instructions; name and email; pickup time	Menu with prices and availability; order total; confirmation number, summary, and pickup time; live order status

Entity	Flows into the system	Flows out of the system
Owner	Login; menu, category, price, and availability changes; order close and cancel; report and search requests	Order history and order details; sales and best-seller reports labeled "Online orders only"; price change history; CSV exports
Kitchen staff (counter included)	Login; ticket status updates (In Progress, Ready)	Kitchen tickets with items, sizes, spice choice, and instructions
Stripe	Verified payment confirmation	Payment request with order amount and order reference
Email service	None	Order-ready request with the customer's email and order number

Two flows run outside the system boundary. Card, Apple Pay, and Google Pay details go from the customer straight to Stripe and never pass through the system (BR-14). The order-ready email goes from the email service to the customer. Refunds happen in Stripe's dashboard, so there is no refund flow between the system and Stripe.

Business rules

The system follows 33 business rules; 28 are approved and 5 rest on proposed requirements that still need a decision. Each rule cites the charter, the RTM IDs, or the team decision of 2 Oct it comes from.

ID	Area	Business rule	Source	Status
BR-01	Ordering	Online orders are for pickup only.	Charter 7.1; F9	Approved
BR-02	Ordering	Orders can be placed only Monday to Saturday, 11:00 to 21:00 Central. Outside those hours the menu stays viewable and ordering is disabled with a message.	F9, 4.17, 4.18	Approved

ID	Area	Business rule	Source	Status
BR-03	Ordering	The pickup time must fall within business hours and be at least the lead time ahead: 15 minutes for regular items, 40 minutes for catering trays.	F9; decision 2 Oct	Approved
BR-04	Ordering	Items the owner marks unavailable are shown as unavailable and cannot be added to the cart.	F2, F27	Approved
BR-05	Ordering	Sized items charge the price of the chosen size: fried rice small or large; catering trays small, medium, or large.	F3	Approved
BR-06	Ordering	The Combination Plate (G1) is two half orders chosen from G2 to G10, sold as one line at the combination price.	F4	Approved
BR-07	Ordering	Any entree can be ordered spicy, and the preference is printed on the kitchen ticket.	F5	Approved
BR-08	Ordering	Customers can add free-text special instructions to an order line.	F6	Proposed
BR-09	Ordering	Customers check out as guests with name and email. No customer accounts exist and no other personal data is collected.	F8, 7.01; updated to-be	Approved
BR-10	Pricing	The system calculates subtotal, tax at 8.25% (Texas 6.25% plus 2% local), and total from database prices. Prices sent by the browser are ignored.	F10; decision 2 Oct	Approved

ID	Area	Business rule	Source	Status
BR-11	Pricing	A price change applies at once to new orders and never alters orders already placed.	F25	Approved
BR-12	Pricing	Every price change records the old price, new price, user, and time.	F26, 4.01	Approved
BR-13	Payment	Every online order is paid in full online before it reaches the kitchen, by card, Apple Pay, or Google Pay through Stripe.	F11, F13, F15	Approved
BR-14	Payment	The system never transmits, processes, or stores card data; Stripe's hosted page collects it.	F12, 8.01	Approved
BR-15	Payment	An order counts as paid only when Stripe sends a verified confirmation, and one confirmation creates exactly one paid order.	F13, 4.02	Approved
BR-16	Payment	If payment fails or a service is down, no order is recorded as paid and nothing reaches the kitchen.	4.09, 4.13	Approved
BR-17	Payment	Stripe runs in test mode during the course; turning on live payments is a post-project handover step.	Charter A4	Approved
BR-18	Kitchen	Only paid orders become kitchen tickets, and each appears within 5 seconds of payment confirmation.	F15, 5.01	Approved

ID	Area	Business rule	Source	Status
BR-19	Kitchen	Tickets move in one fixed order, Received to In Progress to Ready, one action per step. A Ready ticket stays Ready until it is closed or cancelled.	F16, 2.05, 2.06	Approved
BR-20	Kitchen	Ticket status shows on the kitchen view, the owner dashboard, and the customer's confirmation page.	F17	Proposed
BR-21	Kitchen	Every ticket status change records the user and time.	4.01	Approved
BR-22	Access	There are two staff roles. Owner sees orders, menu, prices, and reports; Kitchen, which counter staff also use, sees the ticket view only. Customers need no login.	F18, F19; decision 2 Oct	Approved
BR-23	Access	Staff can change their own password.	F20	Proposed
BR-24	Records	Completed orders are kept, never deleted, for at least 12 months.	10.01, 5.02	Proposed
BR-25	Reporting	Reports count paid online orders over a date range the owner picks: total sales, order count, average order value, and best sellers by quantity and revenue. Cancelled orders are excluded.	F28, F29, 10.02; decision 2 Oct	Approved
BR-26	Reporting	Any report can be exported to a CSV file.	F30	Proposed
BR-27	Settings	Business hours, lead times, and the tax rate are stored as settings, not written into code.	3.07; decision 2 Oct	Approved

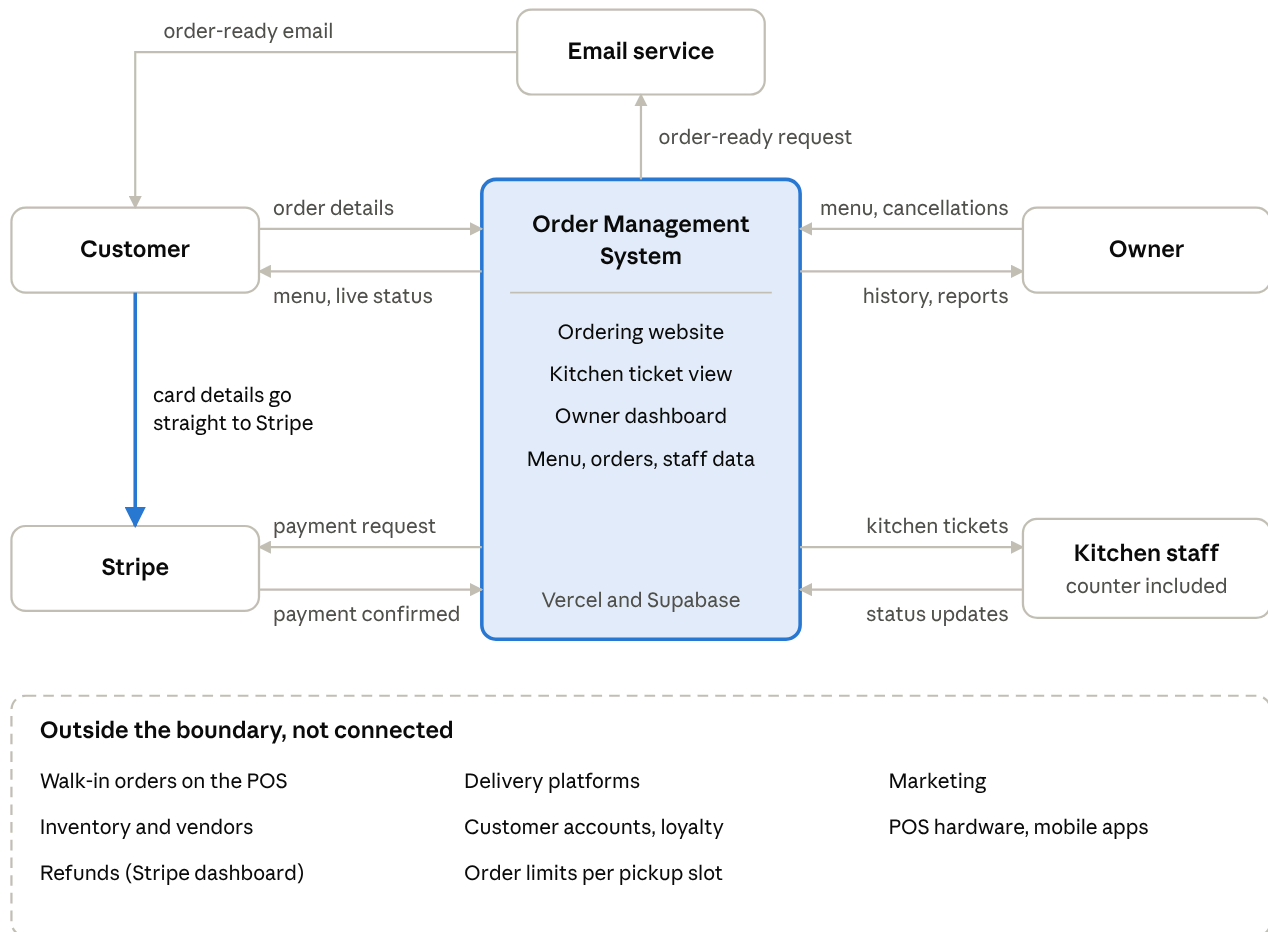
ID	Area	Business rule	Source	Status
BR-28	Notification	When a ticket is marked Ready, the system has the email service send the customer one order-ready email.	Decision 2 Oct; new	Approved
BR-29	Orders	The owner can mark an order cancelled. The system records the status only; it sends nothing to Stripe.	Decision 2 Oct	Approved
BR-30	Orders	Refunds are issued by the owner in Stripe's dashboard, outside the system.	Decision 2 Oct	Approved
BR-31	Orders	An unclaimed order stays Ready until the owner closes or cancels it by hand. Nothing expires automatically.	Decision 2 Oct	Approved
BR-32	Reporting	Every report is labeled "Online orders only" in the interface and the user manual.	Decision 2 Oct	Approved
BR-33	Ordering	Online orders per pickup slot are not capped in this version; this is a known limitation.	Decision 2 Oct	Approved

Scope diagram guide

Draw the system as one central shape with the five external entities around it, every flow from the data flow table as a labeled arrow, and the out-of-scope items outside the boundary. The sketch below is a starting point for Noor to redraw in Visio, the format the course requires.

Online orders flow through one system; card data goes straight to Stripe

Starting sketch for the scope diagram; arrows match the data flow table (updated 2 Oct)



context-level scope diagram sketch · 1 system, 5 external entities, 11 flows (updated 2 Oct)

The card details arrow is the one to get right: it runs from the customer to Stripe and never touches the system. The order-ready email also runs outside the system, from the email service to the customer.

Checklist for the final diagram:

- ☐ One system shape, named "Menciu's Chopstix Order Management System"
- ☐ Five external entities: Customer, Owner, Kitchen staff (counter included), Stripe, Email service
- ☐ Each flow from the data flow table drawn as a labeled arrow in the right direction
- ☐ The card details arrow drawn from Customer to Stripe, outside the system
- ☐ The order-ready email drawn from Email service to Customer, outside the system
- ☐ No refund arrow between the system and Stripe; refunds sit outside the boundary

- ☐ Out-of-scope items placed outside the boundary with no arrows to the system
- ☐ Vercel and Supabase shown inside the system or left off, never as external entities
- ☐ The same five entities used in the DFDs, so the scope diagram and DFDs match
- ☐ Saved as a Visio file and exported to PDF, named Team09-FA2026-CIS4375-ScopeDiagram

Decisions and open items

The team answered eight of the nine questions on 2 Oct. Only the order-ready email changes the diagram, by adding the email service as an external entity.

Question	Decision	Effect on the diagram
Refunds and cancellations	The owner refunds in Stripe's dashboard; the system only marks the order cancelled	None; no refund flow to Stripe
Customer notification	Confirmation page with live status, plus an email when the order is Ready	Adds the email service entity
Counter staff	Same view and same role as the kitchen	One "Kitchen staff" entity
Walk-in sales in reports	Online orders only; every report labeled "Online orders only"	None
Lead time for regular items	15 minutes; catering trays stay at 40; both are settings	None
Tax rate	8.25%, stored as a setting; checked against a receipt before seeding	None
No-shows	Stay Ready until the owner closes or cancels them; no automatic expiry	None
Order limits per slot	No cap in this version; a known limitation	Listed outside the boundary
Proposed rules	Pending approve-or-drop on BR-08, BR-20, BR-23, BR-24, BR-26	None

Still open:

- ☐ Approve or drop BR-08, BR-20, BR-23, BR-24, and BR-26.
- ☐ **Phone number:** the updated to-be diagrams collect name and email, but RTM F8 and 7.01 still say name and phone. Confirm whether the phone number is still collected.
- ☐ **Closing a normal order:** the owner closes no-shows by hand, but nothing says who closes an order once it is picked up. Decide whether kitchen staff close it or the owner does.
- ☐ **Email service:** choose the provider, confirm its free tier fits, and include it with Stripe, Supabase, and Vercel in the vendor approval request to the Program Manager.
- ☐ **RTM updates:** change F8 and 7.01 to email, F9 to the 15-minute lead time, and F16 to include Closed and Cancelled; add requirements for BR-28, BR-29, and BR-32.
- ☐ **Change log:** record the order-ready email as a change, since the charter did not include it.

One wording fix for the charter: it says order intake is handled manually, but the as-is swim lane shows a POS and card reader at the counter. "Taken at the counter and relayed to the kitchen by voice" matches both.